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PHP

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❑What is PHP?

- PHP is an acronym for "PHP: Hypertext Preprocessor"
- PHP is a widely-used, open source scripting language
- PHP scripts are executed on the server
- PHP is free to download and use

- **Server-side scripting runs on the web server**. It processes requests, interacts with databases, and send results to the browser.

- **Client-side scripting runs on the user's browser.** It mainly used to make web pages interactive and dynamic without contacting the server every time**.**

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❑What is a PHP File?

- PHP files can contain text, HTML, CSS, JavaScript, and PHP code
- PHP code is executed on the server, and the result is returned to the browser as plain HTML
- PHP files have extension ".php"

What Can PHP Do?

- PHP can generate dynamic page content
- PHP can create, open, read, write, delete, and close files on the server
- PHP can collect form data
- PHP can send and receive cookies
- PHP can add, delete, modify data in your database
- PHP can be used to control user-access
- PHP can encrypt data

❑Why PHP?

- PHP runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.)
- PHP is compatible with almost all servers used today (Apache, IIS, etc.)
- PHP supports a wide range of databases
- PHP is free. Download it from the official PHP resource: www.php.net
- PHP is easy to learn and runs efficiently on the server side

Basic PHP Syntax

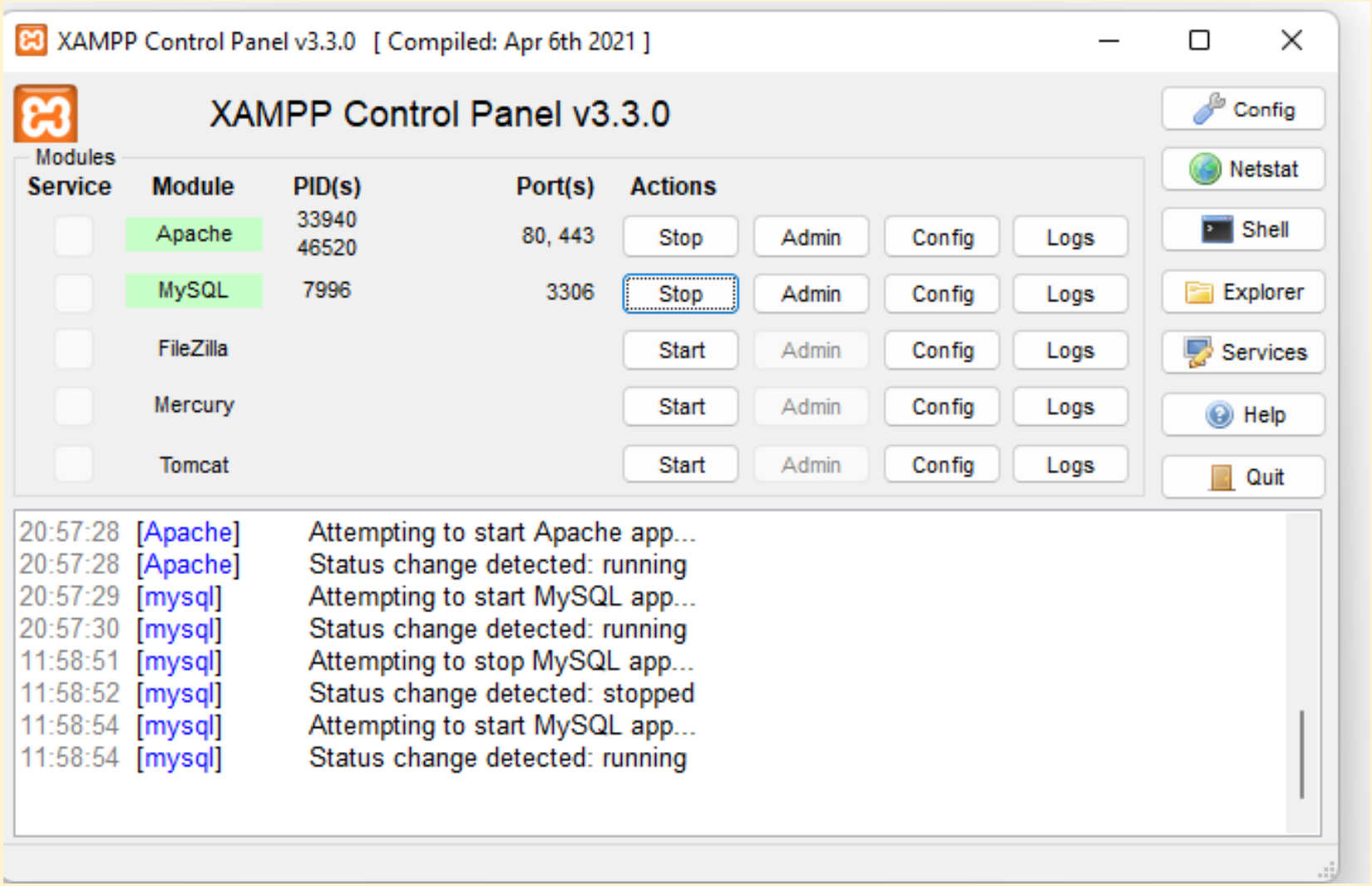
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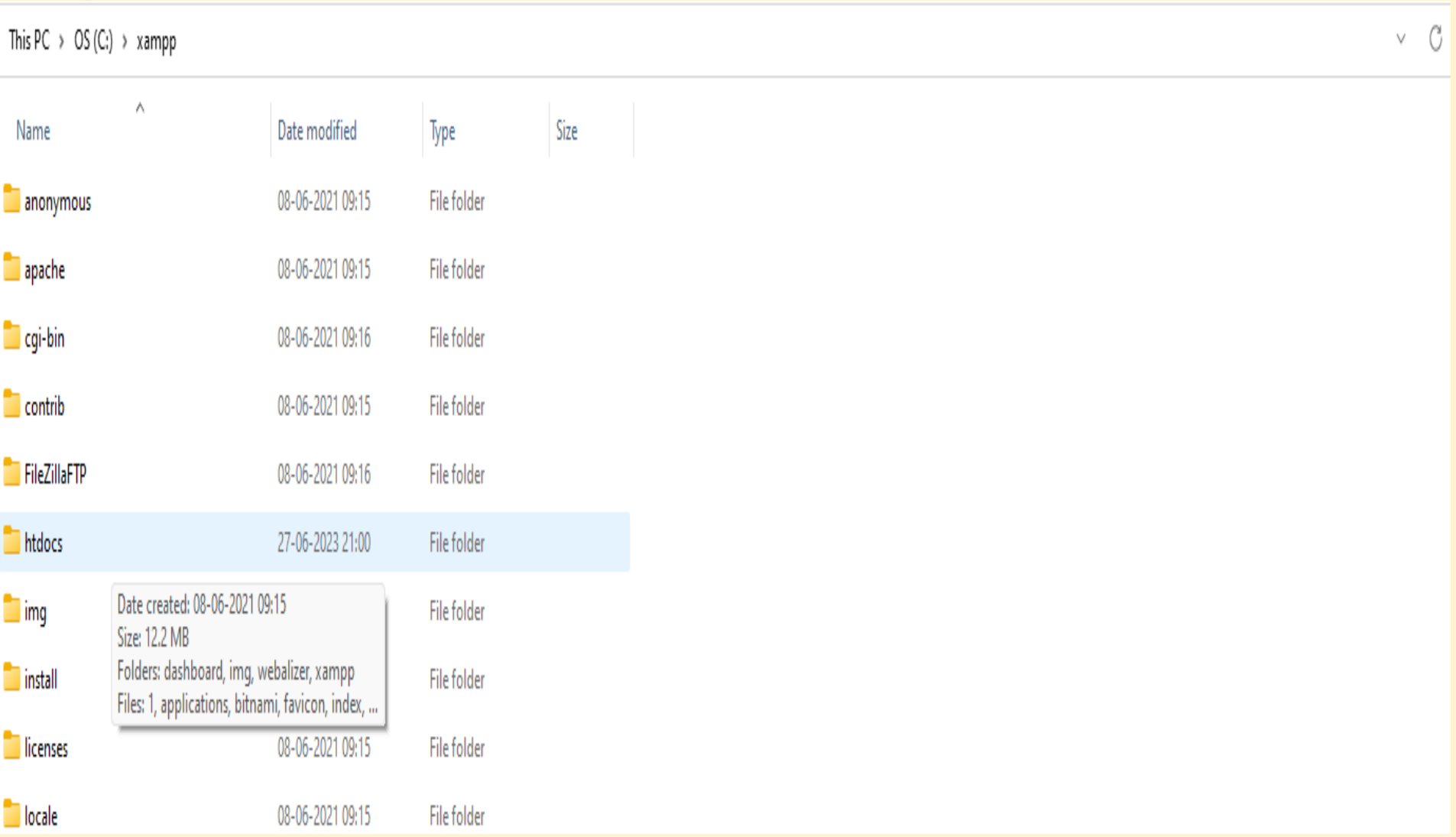
- A PHP script starts with **<?php** and ends with **?>**:

```
<?php
// PHP code goes here
?>
```

- The default file extension for PHP files is ".php".
- A PHP file normally contains HTML tags, and some PHP scripting code.

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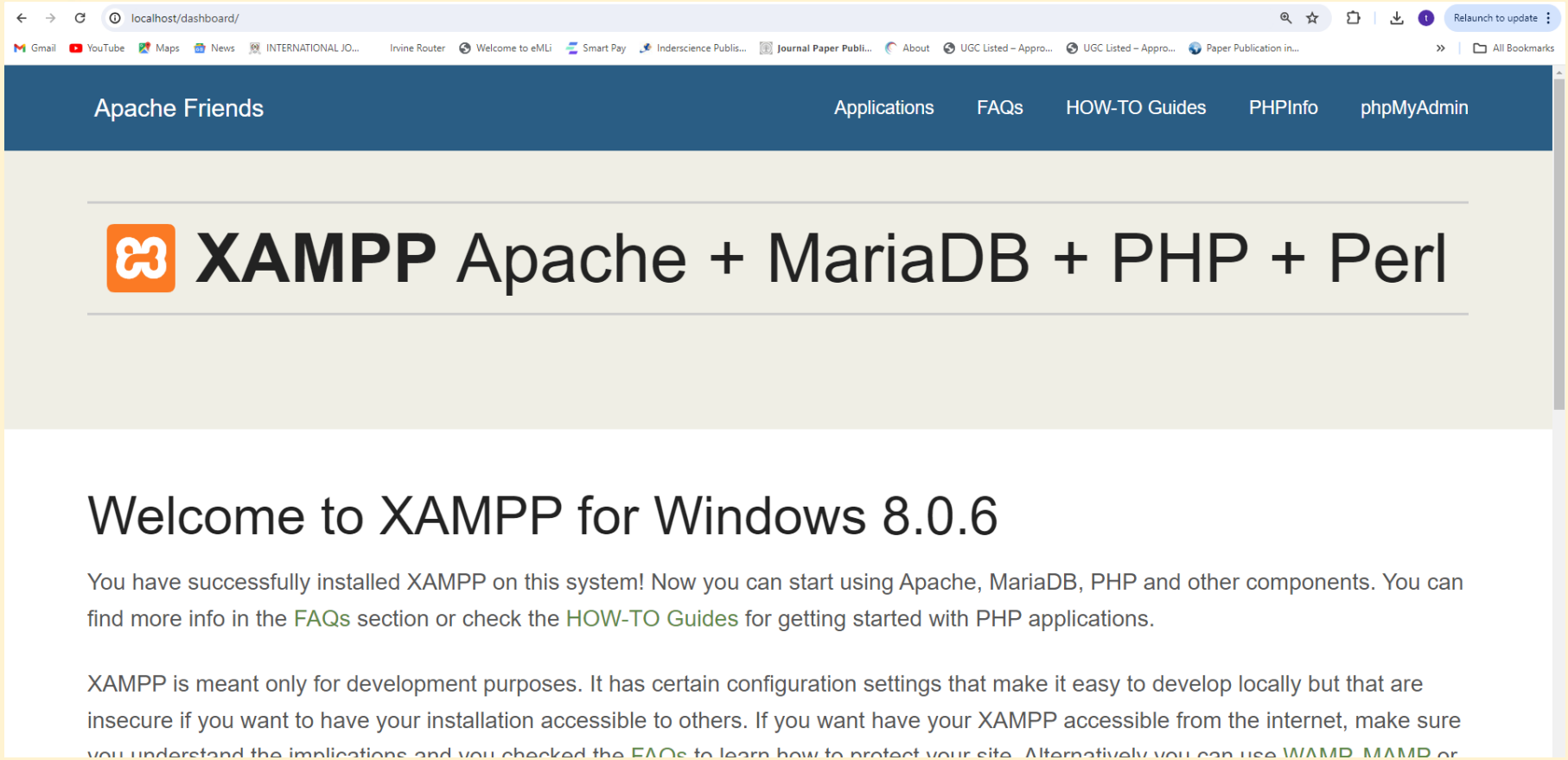

This PC › OS (C:) › xampp
Name
Date modified
Type
Size
anonymous 08-06-2021 09:15 File folder
apache 08-06-2021 09:15 File folder
cgi-bin 08-06-2021 09:16 File folder
contrib 08-06-2021 09:15 File folder
FileZillaFTP 08-06-2021 09:16 File folder
htdocs 27-06-2023 21:00 File folder
img File folder
install File folder
licenses 08-06-2021 09:15 File folder
locale 08-06-2021 09:15 File folder
Date created: 08-06-2021 09:15
Size: 12.2 MB
Folders: dashboard, img, webalizer, xampp
Files: 1, applications, bitnami, favicon, index, ...

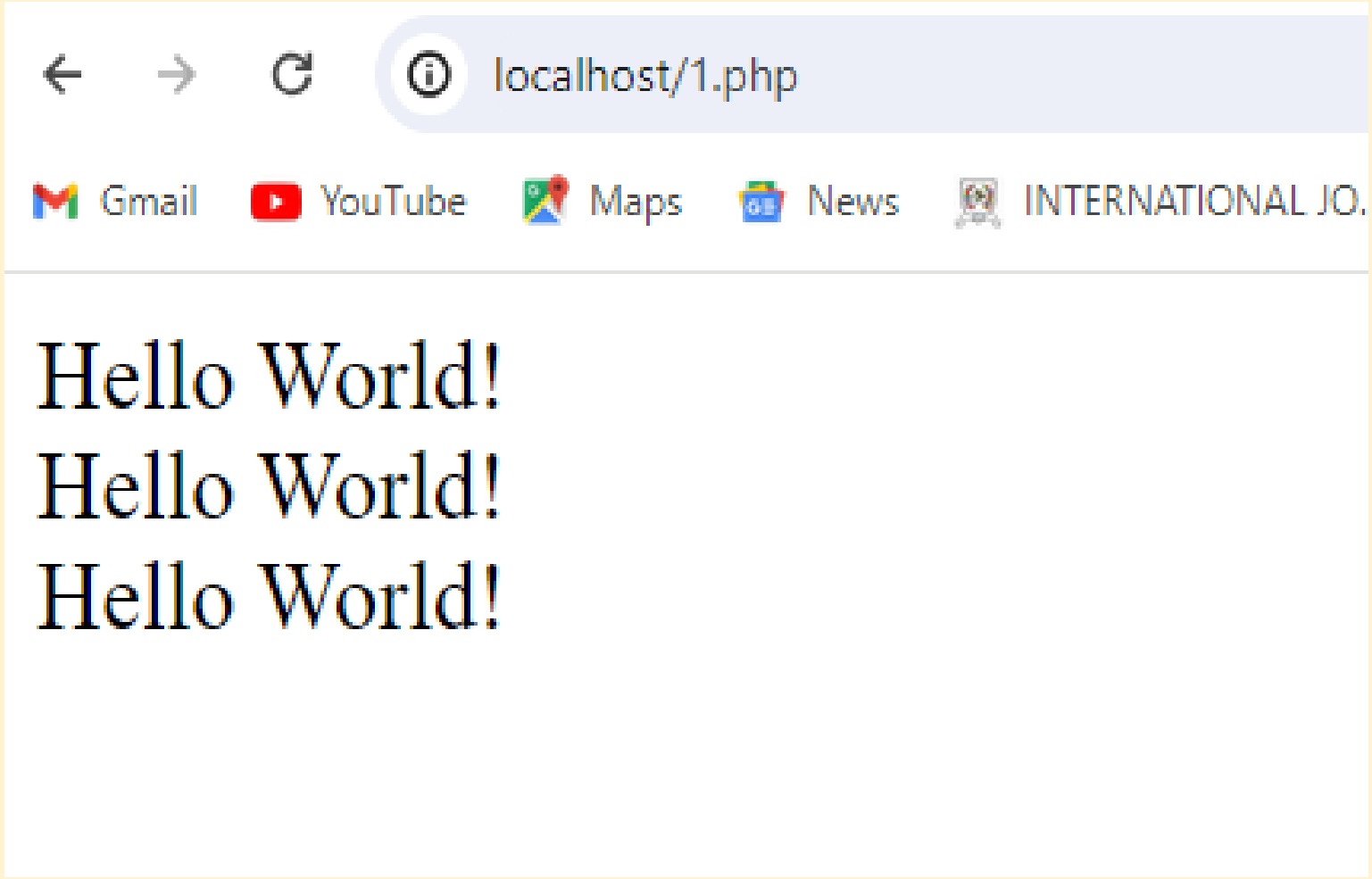
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This PC > OS (C:) > xampp > htdocs >

Name	Date modified	Type	Size
dashboard	08-06-2021 09:15	File folder	
img	08-06-2021 09:15	File folder	
webalizer	08-06-2021 09:15	File folder	
xampp	08-06-2021 09:15	File folder	
1	27-06-2023 21:00	PHP File	1 KB
applications	27-08-2019 19:32	Microsoft Edge H...	4 KB
bitnami	27-08-2019 19:32	Cascading Style S...	1 KB
favicon	16-07-2015 21:02	Icon	31 KB
index	16-07-2015 21:02	PHP File	1 KB
New Text Document (2)	27-06-2023 21:00	Text Document	0 KB
New Text Document	08-06-2021 09:41	Text Document	0 KB
PRAC 13(2)	08-06-2021 10:17	PHP File	1 KB
PRAC 13	08-06-2021 10:12	PHP File	1 KB
PRAC 14	08-06-2021 11:09	File	2 KB
PRAC 14(2)	08-06-2021 11:43	PHP File	1 KB
PRAC 14	08-06-2021 13:14	PHP File	2 KB
PRAC 15	08-06-2021 13:21	PHP File	2 KB
PRAC 16	08-06-2021 13:35	Microsoft Edge H...	2 KB
PRAC 16	08-06-2021 13:34	PHP File	2 KB
PRAC 17(1)	08-06-2021 13:54	PHP File	1 KB
PRAC 17(2)	08-06-2021 13:55	File	1 KB
PRAC 17(2)	08-06-2021 13:56	PHP File	1 KB
retrive	08-06-2021 13:54	PHP File	1 KB
retrivehidden	08-06-2021 13:55	PHP File	1 KB
welcome	08-06-2021 13:35	PHP File	1 KB

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localhost/1.php
Gmail
YouTube
Maps
News
INTERNATIONAL JO.
Hello World!
Hello World!
Hello World!

Example

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```
<!DOCTYPE html>
<html>
<body>

<h1>My first PHP page</h1>

<?php
echo "Hello World!";
?>

</body>
</html>
```

Comments in PHP

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- A comment in PHP code is a line that is not read/executed as part of the program. Its only purpose is to be read by someone who is looking at the code.

```
//  single line comment
# single line comment
/*
Multiple line comment
*/
```

Example

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```
<!DOCTYPE html>
<html>
<body>

<?php
// This is a single-line comment

# This is also a single-line comment

/*
This is a multiple-lines comment block
that spans over multiple
lines
*/

// You can also use comments to leave out parts of a code line
$x = 5 /* + 15 */ + 5;
echo $x;
?>

</body>
```

PHP Case Sensitivity

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- In PHP, all keywords (e.g. if, else, while, echo, etc.), classes, functions, and user-defined functions are NOT case-sensitive.
- Variables and constant names are case-sensitive.

Example

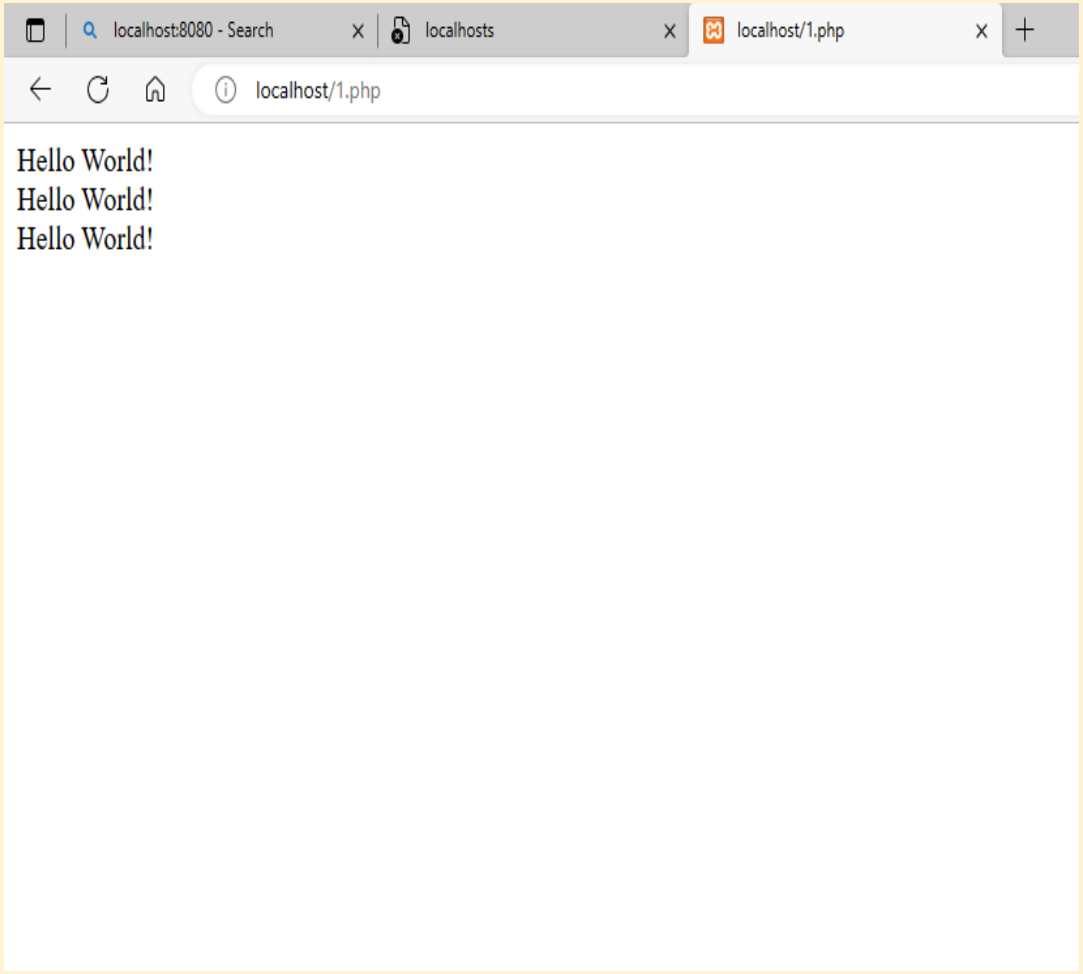
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```
<!DOCTYPE html>
<html>
<body>

<?php
ECHO "Hello World!<br>";
echo "Hello World!<br>";
EcHo "Hello World!<br>";
?>

</body>
</html>
```

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- However; all variable names are case-sensitive.

```
<!DOCTYPE html>
<html>
<body>

<?php
$color = "red";
echo "My car is " . $color . "<br>";
echo "My house is " . $COLOR . "<br>";
echo "My boat is " . $coLOR . "<br>";
?>

</body>
</html>
```

My car is red

My house is

My boat is

PHP Variables

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- Variables are "containers" for storing information.

```php
<?php
$txt = "Hello world!";
$x = 5;
$y = 10.5;
?>
```

```php
<!DOCTYPE html>
<html>
<body>

<?php
$txt = "Hello world!";
$x = 5;
$y = 10.5;

echo $txt;
echo "<br>";
echo $x;
echo "<br>";
echo $y;
?>

</body>
</html>
```

Hello world!
5
10.5

PHP Variables

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- A variable can have a short name (like x and y) or a more descriptive name (age, carname, total_volume).
- Rules for PHP variables:
  - A variable starts with the $ sign, followed by the name of the variable
  - A variable name must start with a letter or the underscore character
  - A variable name cannot start with a number
  - A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
  - Variable names are case-sensitive ($age and $AGE are two different variables)

PHP Echo/Print

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- echo and print are more or less the same. They are both used to output data to the screen.
- The differences are small:
  - echo has no return value while print has a return value of 1.
  - echo can take multiple parameters (although such usage is rare) while print can take one argument.
  - echo is marginally faster than print.

PHP Echo Statement

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- The echo statement can be used with or without parentheses: echo or echo()

Run ›

```
<!DOCTYPE html>
<html>
<body>

<?php
$txt1 = "Learn PHP";
$txt2 = "W3Schools.com";
$x = 5;
$y = 4;

echo "<h2>" . $txt1 . "</h2>";
echo "Study PHP at " . $txt2 . "<br>";
echo $x + $y;
?>

</body>
</html>
```

Learn PHP

Study PHP at W3Schools.com

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The PHP print Statement

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```
<!DOCTYPE html>
<html>
<body>

<?php
$txt1 = "Learn PHP";
$txt2 = "W3Schools.com";
$x = 5;
$y = 4;

echo "<h2>" . $txt1 . "</h2>";
echo "Study PHP at " . $txt2 . "<br>";
echo $x + $y;
?>

</body>
</html>
```

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Study PHP at W3Schools.com

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PHP Data types

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- PHP String

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = "Hello world!";
$y = 'Hello world!';

echo $x;
echo "<br>";
echo $y;
?>

</body>
</html>
```

Hello world!

Hello world!

PHP Data types

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- **PHP Integer**
- The PHP var_dump() function returns the data type and value:

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 5985;
var_dump($x);
?>

</body>
</html>
```

```
int(5985)
```

PHP Data types

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- **PHP Float**
- The PHP var_dump() function returns the data type and value:

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 10.365;
var_dump($x);
?>

</body>
</html>
```

```
float(10.365)
```

PHP Data types

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- **PHP Array**
- The PHP var_dump() function returns the data type and value:

```
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array("Volvo","BMW","Toyota");
var_dump($cars);
?>

</body>
</html>
```

```
array(3) {
  [0]=>
  string(5) "Volvo"
  [1]=>
  string(3) "BMW"
  [2]=>
  string(6) "Toyota"
}
```

- Print two variables in single echo

```
<?php
$message_1 = "Good Morning.";
$message_2 = "Have a nice day!";
echo $message_1." ". $message_2;
?>
```

PHP String Functions

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- strlen() - Return the Length of a String

```
<!DOCTYPE html>
<html>
<body>

<?php
echo strlen("Hello world!");
?>

</body>
</html>
```

```
12
```

PHP String Functions

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- str_word_count() - Count Words in a String

```
<!DOCTYPE html>
<html>
<body>

<?php
echo str_word_count("Hello world!");
?>

</body>
</html>
```

```
2
```

PHP String Functions

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- strrev() - Reverse a String

```
<!DOCTYPE html>
<html>
<body>

<?php
echo strrev("Hello world!");
?>

</body>
</html>
```

```
!dlrow olleH
```

PHP String Functions

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- str_replace() - Replace Text Within a String

```
<!DOCTYPE html>
<html>
<body>

<?php
echo str_replace("world", "Dolly", "Hello world!");
?>

</body>
</html>
```

Hello Dolly!

Excercise

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- Write a program to print your biodata using echo.
- Write a program to print your biodata using variable.
- Write a program to print a string using echo+variable.
- Write a program to print two variables in single echo

PHP Operators

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- Operators are used to perform operations on variables and values.
- PHP divides the operators in the following groups:
  - Arithmetic operators
  - Assignment operators
  - Comparison operators
  - Increment/Decrement operators
  - Logical operators
  - String operators
  - Array operators

PHP Arithmetic Operator

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Operator	Name	Example	Result
+	Addition	$x + $y	Sum of $x and $y
-	Subtraction	$x - $y	Difference of $x and $y
*	Multiplication	$x * $y	Product of $x and $y
/	Division	$x / $y	Quotient of $x and $y
%	Modulus	$x % $y	Remainder of $x divided by $y
**	Exponentiation	$x ** $y	Result of raising $x to the $y'th power (Introduced in PHP 5.6)

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 10;
$y = 3;

echo $x ** $y;
?>

</body>
</html>
```

1000

PHP Assignment Operators

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Assignment	Same as...	Description
x = y	x = y	The left operand gets set to the value of the expression on the right
x += y	x = x + y	Addition
x -= y	x = x - y	Subtraction
x *= y	x = x * y	Multiplication
x /= y	x = x / y	Division
x %= y	x = x % y	Modulus

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 10;
$x /= 5;

echo $x;
?>

</body>
</html>
```

2

PHP Comparison Operators

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Operator	Name	Example	Result
==	Equal	$x == $y	Returns true if $x is equal to $y
===	Identical	$x === $y	Returns true if $x is equal to $y, and they are of the same type
!=	Not equal	$x != $y	Returns true if $x is not equal to $y
<>	Not equal	$x <> $y	Returns true if $x is not equal to $y
!==	Not identical	$x !== $y	Returns true if $x is not equal to $y, or they are not of the same type
>	Greater than	$x > $y	Returns true if $x is greater than $y
<	Less than	$x < $y	Returns true if $x is less than $y
>=	Greater than or equal to	$x >= $y	Returns true if $x is greater than or equal to $y
<=	Less than or equal to	$x <= $y	Returns true if $x is less than or equal to $y

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;
$y = "100";

var_dump($x <> $y); // returns false because values are equal
?>

</body>
</html>
```

```
bool(false)
```

PHP Increment / Decrement Operators

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Operator	Name	Description
++$x	Pre-increment	Increments $x by one, then returns $x
$x++	Post-increment	Returns $x, then increments $x by one
--$x	Pre-decrement	Decrements $x by one, then returns $x
$x--	Post-decrement	Returns $x, then decrements $x by one

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 10;
echo ++$x;
?>

</body>
</html>
```

```
11
```

PHP Logical Operators

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Operator	Name	Example	Result
and	And	$x and $y	True if both $x and $y are true
or	Or	$x or $y	True if either $x or $y is true
xor	Xor	$x xor $y	True if either $x or $y is true, but not both
&&	And	$x && $y	True if both $x and $y are true
||	Or	$x || $y	True if either $x or $y is true
!	Not	!$x	True if $x is not true

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;
$y = 50;

if ($x == 100 and $y == 50) {
    echo "Hello world!";
}
?>

</body>
</html>
```

Hello world!

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;
$y = 50;

if ($x == 10 and $y == 50) {
    echo "Hello world!";
}
?>

</body>
</html>
```

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;
$y = 50;

if ($x == 100 xor $y == 80) {
    echo "Hello world!";
}
?>

</body>
</html>
```

Hello world!

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;
$y = 50;

if ($x == 100 xor $y == 50) {
    echo "Hello world!";
}
?>

</body>
</html>
```

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 100;

if ($x !== 90) {
    echo "Hello world!";
}
?>

</body>
</html>
```

Hello world!

PHP Strings

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- A string is a sequence of characters, like "Hello world!".

PHP String Functions

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- Get The Length of a String
- The PHP strlen() function returns the length of a string.

```php
<?php
echo strlen("Hello world!"); // outputs 12
?>
```

Count The Number of Words in a String

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- The PHP str_word_count() function counts the number of words in a string:

```php
<?php
echo str_word_count("Hello world!"); // outputs 2
?>
```

Reverse a String

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- The PHP strrev() function reverses a string:

```php
<?php
echo strrev("Hello world!"); // outputs !dlrow olleH
?>
```

Search For a Specific Text Within a String

- The PHP strpos() function searches for a specific text within a string.
- If a match is found, the function returns the character position of the first match. If no match is found, it will return FALSE.

```php
<?php
echo strpos("Hello world!", "world"); // outputs 6
?>
```

Replace Text Within a String

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- The PHP str_replace() function replaces some characters with some other characters in a string

```php
<?php
echo str_replace("world", "Dolly", "Hello world!"); // outputs Hello Dolly!
?>
```

PHP Conditional Statements



- Very often when you write code, you want to perform different actions for different conditions. You can use conditional statements in your code to do this.
- In PHP we have the following conditional statements:
  - if statement - executes some code if one condition is true
  - if...else statement - executes some code if a condition is true and another code if that condition is false
  - if...elseif....else statement - executes different codes for more than two conditions
  - switch statement - selects one of many blocks of code to be executed

PHP - The if Statement

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```
if (condition) {
    code to be executed if condition is true;
}
```

```
<!DOCTYPE html>
<html>
<body>

<?php
if (5 > 3) {
  echo "Have a good day!";
}
?>

</body>
</html>
```

Output: "Have a good day!"

PHP - The if...else Statement

```
if (condition) {
    code to be executed if condition is true;
} else {
    code to be executed if condition is false;
}
```

```php
<?php
$t = date("H");

if ($t < "20") {
    echo "Have a good day!";
} else {
    echo "Have a good night!";
}
?>
```

Output: "Have a good day!" if the current time is less than 20, and "Have a good night!" otherwise

PHP - The if...elseif....else Statement

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```
if (condition) {
    code to be executed if this condition is true;
} elseif (condition) {
    code to be executed if this condition is true;
} else {
    code to be executed if all conditions are false;
}
```

PHP - The if...elseif....else Statement

```php
<?php
$t = date("H");

if ($t < "10") {
    echo "Have a good morning!";
} elseif ($t < "20") {
    echo "Have a good day!";
} else {
    echo "Have a good night!";
}
?>
```

Output: "Have a good morning!" if the current time is less than 10, and "Have a good day!" if the current time is less than 20. Otherwise it will output "Have a good night!"

PHP Switch Statement

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- The switch statement is used to perform different actions based on different conditions.

```
switch (n) {
    case label1:
        code to be executed if n=label1;
        break;
    case label2:
        code to be executed if n=label2;
        break;
    case label3:
        code to be executed if n=label3;
        break;
    ...
    default:
        code to be executed if n is different from all labels;
}
```

PHP Switch Statement

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```php
<?php
$favcolor = "red";

switch ($favcolor) {
    case "red":
        echo "Your favorite color is red!";
        break;
    case "blue":
        echo "Your favorite color is blue!";
        break;
    case "green":
        echo "Your favorite color is green!";
        break;
    default:
        echo "Your favorite color is neither red, blue, nor green!";
}
?>
```

Exercise

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- Create a form using two text fields and submit button.
- Retrieve the both values in php script.
- Use the if statement to get the biggest number among them.
- Display the biggest number.

Exercise

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- Create a resume using at least 10 fields.
- Display the form values in php script.

Exercise

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- Create a form to take following values from user: Name, Marks of 5 subjects and submit button.
- Retrieve the form values on PHP script.
- Display the student Name, Marks of 5 subjects, Grade, Percentage and Total in Table form.

PHP Loops

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- Often when you write code, you want the same block of code to run over and over again in a row. Instead of adding several almost equal code-lines in a script, we can use loops to perform a task like this.
- In PHP, we have the following looping statements:
  - while - loops through a block of code as long as the specified condition is true
  - do...while - loops through a block of code once, and then repeats the loop as long as the specified condition is true
  - for - loops through a block of code a specified number of times

The PHP while Loop

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```
while (condition is true) {
    code to be executed;
}
```

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 1;

while($x <= 5) {
  echo "The number is: $x <br>";
  $x++;
}
?>

</body>
</html>
```

The number is: 1
The number is: 2
The number is: 3
The number is: 4
The number is: 5

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 0;

while($x <= 100) {
  echo "The number is: $x <br>";
  $x+=10;
}
?>

</body>
</html>
```

The number is: 0

The number is: 10

The number is: 20

The number is: 30

The number is: 40

The number is: 50

The number is: 60

The number is: 70

The number is: 80

The number is: 90

The number is: 100

The PHP do...while Loop

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```
do {
    code to be executed;
} while (condition is true);
```

Note: In a do...while loop the condition is tested AFTER executing the statements within the loop. This means that the do...while loop will execute its statements at least once, even if the condition is false.

```php
<!DOCTYPE html>
<html>
<body>

<?php
$x = 1;

do {
  echo "The number is: $x <br>";
  $x++;
} while ($x <= 5);
?>

</body>
</html>
```

The number is: 1

The number is: 2

The number is: 3

The number is: 4

The number is: 5

```
<!DOCTYPE html>
<html>
<body>

<?php
$x = 6;

do {
  echo "The number is: $x <br>";
  $x++;
} while ($x <= 5);
?>

</body>
</html>
```

The number is: 6

PHP for loop

```
for (init counter; test counter; increment counter) {
    code to be executed;
}
```

```
<!DOCTYPE html>
<html>
<body>

<?php
for ($x = 0; $x <= 10; $x++) {
  echo "The number is: $x <br>";
}
?>

</body>
</html>
```

The number is: 0
The number is: 1
The number is: 2
The number is: 3
The number is: 4
The number is: 5
The number is: 6
The number is: 7
The number is: 8
The number is: 9
The number is: 10

```php
<!DOCTYPE html>
<html>
<body>

<?php
for ($x = 0; $x <= 100; $x+=10) {
  echo "The number is: $x <br>";
}
?>

</body>
</html>
```

The number is: 0
The number is: 10
The number is: 20
The number is: 30
The number is: 40
The number is: 50
The number is: 60
The number is: 70
The number is: 80
The number is: 90
The number is: 100

PHP Break and Continue

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- The break statement can also be used to jump out of a loop.

```
<!DOCTYPE html>
<html>
<body>

<?php
for ($x = 0; $x < 10; $x++) {
  if ($x == 4) {
    break;
  }
  echo "The number is: $x <br>";
}
?>

</body>
</html>
```

The number is: 0
The number is: 1
The number is: 2
The number is: 3

```php
<!DOCTYPE html>
<html>
<body>

<?php
for ($x = 0; $x < 10; $x++) {
  if ($x == 4) {
    continue;
  }
  echo "The number is: $x <br>";
}
?>

</body>
</html>
```

The number is: 0

The number is: 1

The number is: 2

The number is: 3

The number is: 5

The number is: 6

The number is: 7

The number is: 8

The number is: 9

Exercise

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- **Write PHP Script to print Fibonacci series.**

0 1 1 2 3 5 8 13 21 34

Here, 0 + 1 = 1

1 + 1 = 2

3 + 2 = 5

- **Logic:**
- Initializing first and second number as 0 and 1.
- Print first and second number.
- From next number, start your loop. So third number will be the sum of the first two numbers.

Example

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Adding 40 items in Drop-Down Menu.

PHP Arrays

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- An array stores multiple values in one single variable:

If you have a list of items (a list of car names, for example), storing the cars in single variables could look like this:

```
$cars1 = "Volvo";
$cars2 = "BMW";
$cars3 = "Toyota";
```

```php
<?php
$cars = array("Volvo", "BMW", "Toyota");
echo "I like " . $cars[0] . ", " . $cars[1] . " and " . $cars[2] . ".";
?>
```

Create an Array in PHP

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- In PHP, the **array() function** is used to create an array:

  array();

- In PHP, there are three types of arrays:
  - **Indexed arrays** - Arrays with a numeric index
  - **Associative arrays** - Arrays with named keys
  - **Multidimensional arrays** - Arrays containing one or more arrays

PHP Indexed Arrays

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- There are two ways to create indexed arrays:
- The index can be assigned automatically (index always starts at 0), like this:

$cars = array("Volvo", "BMW", "Toyota");

- or the index can be assigned manually:

```
$cars[0] = "Volvo";
$cars[1] = "BMW";
$cars[2] = "Toyota";
```

Get The Length of an Array - The count() Function



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```php
<?php
$cars = array("Volvo", "BMW", "Toyota");
echo count($cars);
?>
```

Example

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```php
<?php
$cars = array("Volvo", "BMW", "Toyota");
echo "I like " . $cars[0] . ", " . $cars[1] . " and " . $cars[2] . ".";
?>
```

Foreach loop

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Syntax

```
foreach ($array as $value) {
  code to be executed;
}
```

For every loop iteration, the value of the current array element is assigned to $value and the array pointer is moved by one, until it reaches the last array element.

```php
<!DOCTYPE html>
<html>
<body>

<?php
$colors = array("red", "green", "blue", "yellow");

foreach ($colors as $value) {
  echo "$value <br>";
}
?>

</body>
</html>
```

red

green

blue

yellow

Array printing using foreach

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```
<?php
$cars=array(“Volvo”,”BMW”,
”Toyoto”);
foreach($cars as $x)
{
        echo $x.”<br>”;
}
?>
```

Array Printing using print_r

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- The print_r() function prints the information about a variable in a more human-readable way.

```
<!DOCTYPE html>
<html>
<body>

<?php
$a = array("red", "green", "blue");
print_r($a);
?>

</body>
</html>
```

```
Array ( [0] => red [1] => green [2] => blue )
```

Loop Through an Indexed Array

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```php
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array("Volvo", "BMW", "Toyota");
$arrlength = count($cars);

for($x = 0; $x < $arrlength; $x++) {
  echo $cars[$x];
  echo "<br>";
}
?>

</body>
</html>
```

Volvo

BMW

Toyota

PHP Associative Arrays

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- Associative arrays are **arrays that use named keys that you assign to them**.
- There are two ways to create an associative array:

```
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
```

- or:

```
$age['Peter'] = "35";
$age['Ben'] = "37";
$age['Joe'] = "43";
```

- The named keys can then be used in a script:

PHP Associative Arrays

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```
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
echo "Peter is " . $age['Peter'] . " years old.";
?>

</body>
</html>
```

Peter is 35 years old.

```
<!DOCTYPE html>
<html>
<body>

<?php
$b = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
print_r($b);
?>

</body>
</html>
```

```
Array ( [Peter] => 35 [Ben] => 37 [Joe] => 43 )
```

Loop Through an Associative Array

- To loop through and print all the values of an associative array, you could use a foreach loop.

The PHP foreach Loop

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- The foreach loop works only on arrays, and is used to loop through each key/value pair in an array.

```
foreach ($array as $value) {
    code to be executed;
}
```

```
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");

foreach($age as $x => $val) {
  echo "$x = $val<br>";
}
?>

</body>
</html>
```

Peter = 35

Ben = 37

Joe = 43

Loop Through an Associative Array

```php
<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");

foreach($age as $x => $x_value) {
    echo "Key=" . $x . ", Value=" . $x_value;
    echo "<br>";
}
?>
```

Key=Peter, Value=35

Key=Ben, Value=37

Key=Joe, Value=43

Example

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```php
<?php
$colors = array("red", "green", "blue", "yellow");

foreach ($colors as $value) {
    echo "$value <br>";
}
?>
```

Example

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```php
<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
echo "Peter is " . $age['Peter'] . " years old.";
?>
```

Multi-dimensional array

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Name	Stock	Sold
Volvo	22	18
BMW	15	13
Saab	5	2
Land Rover	17	15

We can store the data from the table above in a two-dimensional array, like this:

```
$cars = array (
  array("Volvo",22,18),
  array("BMW",15,13),
  array("Saab",5,2),
  array("Land Rover",17,15)
);
```

Multi-dimensional array

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Multidimensional Array or Nested Array :

Emp No.	Name	Designation	Salary
1	Krishana	Manager	50000
2	Salman	Salesman	20000
3	Mohan	Computer Operator	12000
4	Amir	Driver	5000



Multidimensional Array or Nested Array :

Emp No.	Name	Designation	Salary
1	Krishana	Manager	50000
2	Salman	Salesman	20000
3	Mohan	Computer Operator	12000
4	Amir	Driver	5000

```
$emp = array (
    array(1,"Krishana","Manager",50000),
    array(2,"Salman","Salesman",20000),
    array(3,"Mohan","Computer Operator",12000),
    array(4,"Amir","Driver",5000)
);
```



Multidimensional Array in PHP :

```
$emp = [
    [ 1 ,"Krishana“ , "Manager“ , 50000 ],
    [ 2 , "Salman“ , "Salesman“ , 20000 ],
    [ 3 , "Mohan“ , "Computer Operator“ , 12000 ],
    [ 4 , "Amir“ , "Driver“ , 5000 ]
];
```



Multidimensional Array in PHP :

```
$emp = [
        0          1                2              3
0   [ 1 ,"Krishana“ , "Manager“ , 50000 ],

1   [ 2 , "Salman“ , "Salesman“ , 20000 ],

2   [ 3 , "Mohan“ , "Computer Operator“ , 12000 ],

3   [ 4 , "Amir“ , "Driver“ , 5000 ]

];
```



Multidimensional Array in PHP :

```
$emp = [
        0          1                2              3
0   [ 1 ,"Krishana“ , "Manager“ , 50000 ],

1   [ 2 , "Salman“ , "Salesman“ , 20000 ],

2   [ 3 , "Mohan“ , "Computer Operator“ , 12000 ],

3   [ 4 , "Amir“ , "Driver“ , 5000 ]

];

    echo $emp[1][1];
```

array.php

```php
<?php
$emp = [
        [1,"Krishana","Manager",50000],
        [2,"Salman","Salesman",20000],
        [3,"Mohan","Computer Operator",12000
        [4,"Amir","Driver",5000]
    ];

echo "<pre>";
print_r($emp);
echo "</pre>";
?>
```

array.php 11:9 CRLF UTF-8 PHP GitHub Git (0)

localhost/LearnPHP/array.php Paused

```
Array
(
    [0] => Array
        (
            [0] => 1
            [1] => Krishana
            [2] => Manager
            [3] => 50000
        )

    [1] => Array
        (
            [0] => 2
            [1] => Salman
            [2] => Salesman
            [3] => 20000
        )

    [2] => Array
        (
            [0] => 3
            [1] => Mohan
            [2] => Computer Operator
            [3] => 12000
        )
```

array.php

```php
<?php
$emp = [
        [1,"Krishana","Manager",50000],
        [2,"Salman","Salesman",20000],
        [3,"Mohan","Computer Operator",1200
        [4,"Amir","Driver",5000]
    ];

echo $emp[0][0] . " ";
echo $emp[0][1] . " ";
echo $emp[0][2] . " ";
echo $emp[0][3] . " ";

echo "<pre>";
print_r($emp);
echo "</pre>";
```

array.php 12:23 CRLF UTF-8 PHP GitHub Git (0)

localhost/LearnPHP/array.php Paused

1 Krishana Manager 50000

```
Array
(
    [0] => Array
        (
            [0] => 1
            [1] => Krishana
            [2] => Manager
            [3] => 50000
        )

    [1] => Array
        (
            [0] => 2
            [1] => Salman
            [2] => Salesman
            [3] => 20000
        )

    [2] => Array
        (
            [0] => 3
            [1] => Mohan
            [2] => Computer Operator
```

array.php

```
        [3,"Mohan","Computer Operator",1200
        [4,"Amir","Driver",5000]
    ];

echo $emp[0][0] . " ";
echo $emp[0][1] . " ";
echo $emp[0][2] . " ";
echo $emp[0][3] . " ";
echo "<br>";
echo $emp[1][0] . " ";
echo $emp[1][1] . " ";
echo $emp[1][2] . " ";
echo $emp[1][3] . " ";

echo "<pre>";
print_r($emp);
```

0 array.php 13:13 CRLF UTF-8 PHP GitHub Git (0)

localhost/LearnPHP/array.php Paused

1 Krishana Manager 50000
2 Salman Salesman 20000

```
Array
(
    [0] => Array
        (
            [0] => 1
            [1] => Krishana
            [2] => Manager
            [3] => 50000
        )

    [1] => Array
        (
            [0] => 2
            [1] => Salman
            [2] => Salesman
            [3] => 20000
        )

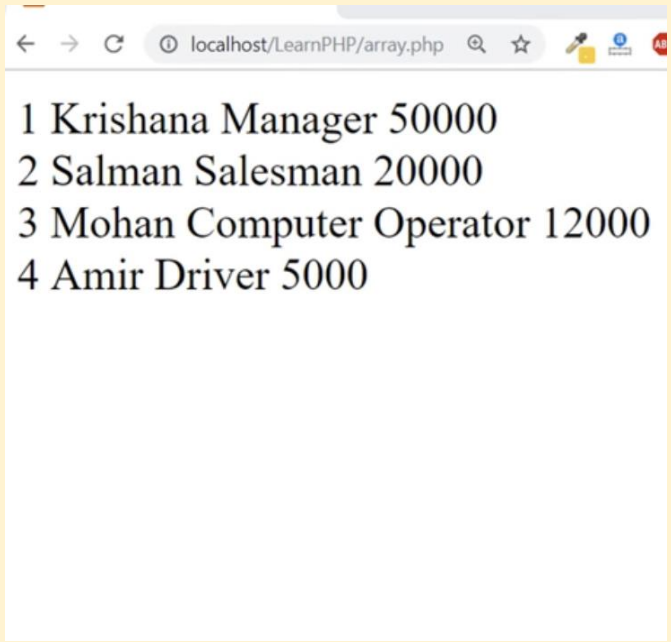
    [2] => Array
        (
            [0] => 3
            [1] => Mohan
```

array.php

```php
        [1,"Krishana","Manager",50000],
        [2,"Salman","Salesman",20000],
        [3,"Mohan","Computer Operator",1200
        [4,"Amir","Driver",5000]
    ];

for($row = 0; $row < 4; $row++){
  for($col = 0; $col < 4; $col++){
    echo $emp[$row][$col] . " ";
  }
  echo "<br>";
}

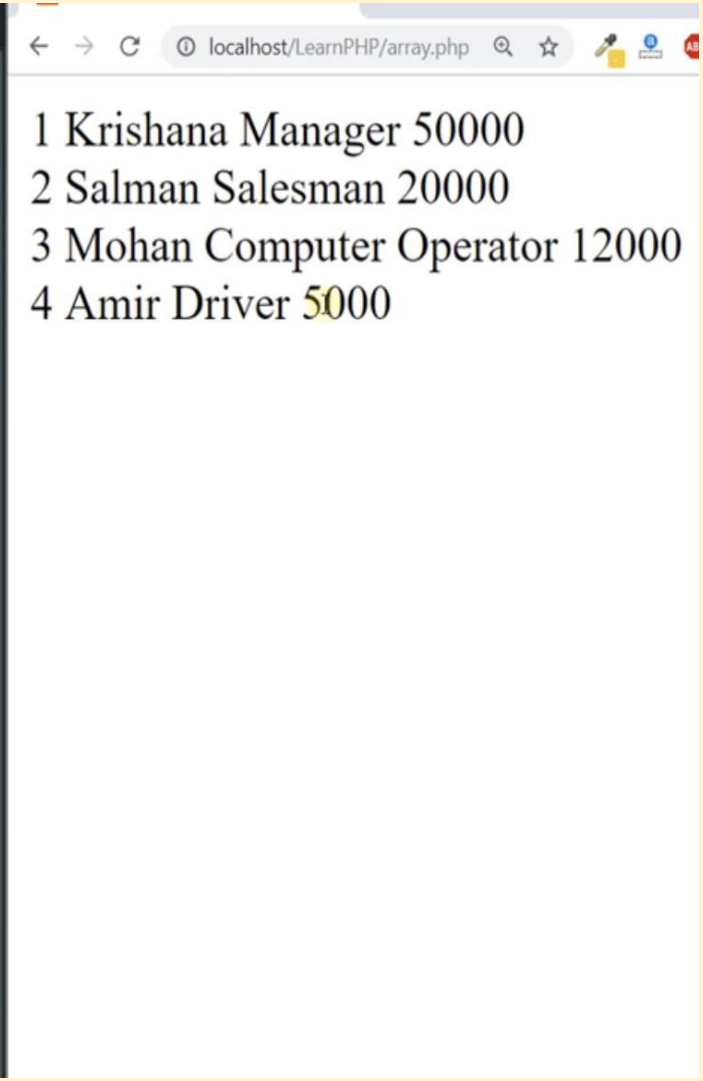

?>
```



array.php

```php
<?php
$emp = [
        [1,"Krishana","Manager",50000],
        [2,"Salman","Salesman",20000],
        [3,"Mohan","Computer Operator",1200
        [4,"Amir","Driver",5000]
    ];

foreach($emp as $v1){
    foreach($v1 as $v2){
        echo $v2 . " ";
    }
    echo "<br>";
}
```



array.php

```
        [1,"Krishana","Manager",50000],
        [2,"Salman","Salesman",20000],
        [3,"Mohan","Computer Operator",1200
        [4,"Amir","Driver",5000]
    ];

echo "<table border='2px'>";
foreach($emp as $v1){
  echo "<tr>";
  foreach($v1 as $v2){
    echo "<td> $v2 </td>";
  }
  echo "</tr>";
}
echo "</table>";
```

localhost/LearnPHP/array.php

1	Krishana	Manager	50000
2	Salman	Salesman	20000
3	Mohan	Computer Operator	12000
4	Amir	Driver	5000

```php
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array (
  array("Volvo",22,18),
  array("BMW",15,13),
  array("Saab",5,2),
  array("Land Rover",17,15)
);

for ($row = 0; $row < 4; $row++) {
  echo "<p><b>Row number $row</b></p>";
  echo "<ul>";
  for ($col = 0; $col < 3; $col++) {
    echo "<li>".$cars[$row][$col]."</li>";
  }
  echo "</ul>";
}
?>

</body>
</html>
```

Row number 0

- Volvo
- 22
- 18

Row number 1

- BMW
- 15
- 13

Row number 2

- Saab
- 5
- 2

Row number 3

- Land Rover
- 17
- 15

Exercise

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Volvo: In stock: 22, sold: 18.

BMW: In stock: 15, sold: 13.

Saab: In stock: 5, sold: 2.

Land Rover: In stock: 17, sold: 15.

```
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array (
  array("Volvo",22,18),
  array("BMW",15,13),
  array("Saab",5,2),
  array("Land Rover",17,15)
);

echo $cars[0][0].": In stock: ".$cars[0][1].", sold: ".$cars[0][2].".<br>";
echo $cars[1][0].": In stock: ".$cars[1][1].", sold: ".$cars[1][2].".<br>";
echo $cars[2][0].": In stock: ".$cars[2][1].", sold: ".$cars[2][2].".<br>";
echo $cars[3][0].": In stock: ".$cars[3][1].", sold: ".$cars[3][2].".<br>";
?>

</body>
</html>
```

- **PHP - Sort Functions For Arrays**
- sort() - sort arrays **in ascending order**
- rsort() - sort arrays in **descending order**
- asort() - sort **associative arrays** in ascending order, **according to the value**
- ksort() - sort **associative arrays** in ascending order, **according to the key**
- arsort() - sort **associative arrays** in descending order, **according to the value**
- krsort() - sort **associative arrays** in descending order, **according to the key**

```php
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array("Volvo", "BMW", "Toyota");
sort($cars);

$clength = count($cars);
for($x = 0; $x < $clength; $x++) {
  echo $cars[$x];
  echo "<br>";
}
?>

</body>
</html>
```

BMW

Toyota

Volvo

```php
<!DOCTYPE html>
<html>
<body>

<?php
$numbers = array(4, 6, 2, 22, 11);
sort($numbers);

$arrlength = count($numbers);
for($x = 0; $x < $arrlength; $x++) {
  echo $numbers[$x];
  echo "<br>";
}
?>

</body>
</html>
```

```
2
4
6
11
22
```

```php
<!DOCTYPE html>
<html>
<body>

<?php
$cars = array("Volvo", "BMW", "Toyota");
rsort($cars);

$clength = count($cars);
for($x = 0; $x < $clength; $x++) {
  echo $cars[$x];
  echo "<br>";
}
?>

</body>
</html>
```

Volvo

Toyota

BMW

```php
<!DOCTYPE html>
<html>
<body>

<?php
$numbers = array(4, 6, 2, 22, 11);
rsort($numbers);

$arrlength = count($numbers);
for($x = 0; $x < $arrlength; $x++) {
  echo $numbers[$x];
  echo "<br>";
}
?>

</body>
</html>
```

```
22
11
6
4
2
```

```php
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
asort($age);

foreach($age as $x => $x_value) {
  echo "Key=" . $x . ", Value=" . $x_value;
  echo "<br>";
}
?>

</body>
</html>
```

Key=Peter, Value=35

Key=Ben, Value=37

Key=Joe, Value=43

```php
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
ksort($age);

foreach($age as $x => $x_value) {
  echo "Key=" . $x . ", Value=" . $x_value;
  echo "<br>";
}
?>

</body>
</html>
```

Key=Ben, Value=37

Key=Joe, Value=43

Key=Peter, Value=35

```php
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
arsort($age);

foreach($age as $x => $x_value) {
  echo "Key=" . $x . ", Value=" . $x_value;
  echo "<br>";
}
?>

</body>
</html>
```

Key=Joe, Value=43

Key=Ben, Value=37

Key=Peter, Value=35

```php
<!DOCTYPE html>
<html>
<body>

<?php
$age = array("Peter"=>"35", "Ben"=>"37", "Joe"=>"43");
krsort($age);

foreach($age as $x => $x_value) {
  echo "Key=" . $x . ", Value=" . $x_value;
  echo "<br>";
}
?>

</body>
</html>
```

Key=Peter, Value=35

Key=Joe, Value=43

Key=Ben, Value=37

Deleting Array Element

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```
unset($cars[4]);
unset($cars);

var_dump($cars); //returning length of array with each element and its length.
```

PHP Function

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- Functions are used to separate out sections of code that perform a particular task.
  - Creating a PHP Function
  - Calling a PHP Function
- Already more than 1000 of built-in library functions created for different area and you just need to call them according to your requirement.

Creating a PHP Function

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```
<?php /* Defining a PHP Function */
    function writeMessage()
    {
        echo "You are really a nice person, Have a
   nice time!";
    }
/* Calling a PHP Function */
     writeMessage();
?>
```

PHP User Defined Functions

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- Besides the built-in PHP functions, it is possible to create your own functions.

Create a User Defined Function in PHP

A user-defined function declaration starts with the word `function`:

Syntax

```
function functionName() {
  code to be executed;
}
```

PHP Functions with Parameters

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```
<?php
        function addFunction($num1, $num2)
         {
              $sum = $num1 + $num2;
              echo "Sum of the two numbers is : $sum";
          }
         addFunction(10, 20);
?>
```

PHP Functions returning value

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```
<?php
    function addFunction($num1, $num2)
  {
                    $sum = $num1 + $num2;
                     return $sum;
   }
 $return_value = addFunction(10, 20);
echo "Returned value from the function:” $return_value";
?>
```

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```
<!DOCTYPE html>
<html>
<body>

<?php
function php() {
    echo "Hello world!";
}

php();
?>

</body>
</html>
```

- Output: Hello world!

PHP Function Arguments

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```
<!DOCTYPE html>
<html>
<body>

<?php
function familyName($fname) {
  echo "$fname Patel.<br>";
}

familyName("Twisha");
familyName("Priya");
familyName("Jeel");
familyName("Jyot");
familyName("Jaivik");
?>

</body>
</html>
```

Twisha Patel.
Priya Patel.
Jeel Patel.
Jyot Patel.
Jaivik Patel.

```php
<!DOCTYPE html>
<html>
<body>

<?php
function familyName($fname, $year) {
  echo "$fname Patel. Born in $year <br>";
}

familyName("Twisha","1993");
familyName("Priya","1995");
familyName("Jeel","1995");
?>

</body>
</html>
```

Twisha Patel. Born in 1993
Priya Patel. Born in 1995
Jeel Patel. Born in 1995

Common PHP String Functions

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Function	What It Does
`empty()`	Tests if a string is empty
`strlen()`	Calculates the number of characters in a string
`strrev()`	Reverses a string
`str_repeat()`	Repeats a string
`substr()`	Retrieves a section of a string
`strcmp()`	Compares two strings
`str_word_count()`	Calculates the number of words in a string
`str_replace()`	Replaces parts of a string
`trim()`	Removes leading and trailing whitespace from a string
`strtolower()`	Lowercases a string
`strtoupper()`	Uppercases a string
`ucfirst()`	Uppercases the first character of a string
`ucwords()`	Uppercases the first character of every word of a string
`addslashes()`	Escapes special characters in a string with backslashes
`stripslashes()`	Removes backslashes from a string
`htmlentities()`	Encodes HTML within a string
`htmlspecialchars()`	Encodes special HTML characters within a string
`nl2br()`	Replaces line breaks in a string with ` ` elements
`html_entity_decode()`	Decodes HTML entities within a string
`htmlspecialchars_decode()`	Decodes special HTML characters within a string
`strip_tags()`	Removes PHP and HTML code from a string

Table 3-3 Common PHP String Functions

Common PHP Numeric Functions

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Function	What It Does
`ceil()`	Rounds a number up
`floor()`	Rounds a number down
`abs()`	Finds the absolute value of a number
`pow()`	Raises one number to the power of another
`log()`	Finds the logarithm of a number
`exp()`	Finds the exponent of a number
`rand()`	Generates a random number
`bindec()`	Converts a number from binary to decimal
`decbin()`	Converts a number from decimal to binary
`decoct()`	Converts a number from decimal to octal
`dechex()`	Converts a number from decimal to hexadecimal
`hexdec()`	Converts a number from hexadecimal to decimal
`octdec()`	Converts a number from octal to decimal
`number_format()`	Formats a number with grouped thousands and decimals
`printf()`	Formats a number using a custom specification

Table 3-4 Common PHP Numeric Functions

Common PHP Array Functions

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Function	What It Does
`explode()`	Splits a string into array elements
`implode()`	Joins array elements into a string
`range()`	Generates a number range as an array
`min()`	Finds the smallest value in an array
`max()`	Finds the largest value in an array
`shuffle()`	Randomly rearranges the sequence of elements in an array
`array_slice()`	Extracts a segment of an array
`array_shift()`	Removes an element from the beginning of an array
`array_unshift()`	Adds an element to the beginning of an array
`array_pop()`	Removes an element from the end of an array
`array_push()`	Adds an element to the end of an array
`array_unique()`	Removes duplicate elements from an array
`array_reverse()`	Reverses the sequence of elements in an array
`array_merge()`	Combines two or more arrays
`array_intersect()`	Calculates the common elements between two or more arrays
`array_diff()`	Calculates the difference between two arrays
`in_array()`	Checks if a particular value exists in an array
`array_key_exists()`	Checks if a particular key exists in an array
`sort()`	Sorts an array
`asort()`	Sorts an associative array by value
`ksort()`	Sorts an associative array by key
`rsort()`	Reverse-sorts an array
`krsort()`	Reverse-sorts an associative array by value
`arsort()`	Reverse-sorts an associative array by key

Table 4-1 Common PHP Array Functions

Array Functions

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1. array_change_key_case(array,CASE_LOWER);
2. array_chunk(array,no);
3. array_column(array,column,column1,…)
4. array_combine(array1,array2)
5. array_diff(array1,array2)
6. array_diff_assoc
7. array_diff_key
8. array_fill(startingindex,length,value)
9. Count
10. Sort
11. asort
12. ksort
13. krsort
14. array_sum
15. array_product
16. array_push
17. array_pop

Array Functions



18. array_fill_keys
19. array_flip
20. array_intersect
21. array_intersect_assoc
22. array_intersect_keys
23. array_key_exists
24. array_keys
25. Shuffle
26. List
27. array_merge
28. array_merge_recursive
29. array_multisort
30. array_pad

Array Functions

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31. array_rand
32. array_shift
33. array_unshift
34. explode
35. Implode
36. Extract
37. in_array
38. array_search
39. array_slice
40. array_reverse
41. array_values

PHP is a Loosely Typed Language

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- adding the **strict declaration**, it will throw a "Fatal Error" if the data type mismatches.

Function without using strict

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```php
<?php
function addNumbers(int $a, int $b) {
  return $a + $b;
}
echo addNumbers(5, "5 days");
// since strict is NOT enabled "5 days" is changed to int(5), and it will return 10
?>
```

```
10
```

- To specify strict we need to set declare(strict_types=1);. This must be on the very first line of the PHP file.

```php
<?php declare(strict_types=1); // strict requirement

function addNumbers(int $a, int $b) {
  return $a + $b;
}
echo addNumbers(5, "5 days");
// since strict is enabled and "5 days" is not an integer, an error will be thrown
?>
```

Output

```
PHP Fatal error: Uncaught TypeError: Argument 2 passed to addNumbers() must be of the type integer, string given,
```

PHP Functions - Returning values

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- To let a function **return a value, use the return statement:**

```
<?php declare(strict_types=1); // strict requirement ?>
<!DOCTYPE html>
<html>
<body>

<?php
function sum(int $x, int $y) {
   $z = $x + $y;
   return $z;
}

echo "5 + 10 = " . sum(5,10) . "<br>";
echo "7 + 13 = " . sum(7,13) . "<br>";
echo "2 + 4 = " . sum(2,4);
?>

</body>
</html>
```

```
5 + 10 = 15
7 + 13 = 20
2 + 4 = 6
```

File Handling in PHP

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- File handling is used to **create, read, write, and append data to files**.

Function	Purpose
`fopen()`	Open a file
`fwrite()`	Write to file
`fread()`	Read from file
`fclose()`	Close file
`file_get_contents()`	Read entire file

Example: Write and Read File

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```
<?php
// Open file in write mode
$file = fopen("data.txt", "w");

// Write data
fwrite($file, "Welcome to PHP File Handling");

// Close file
fclose($file);

// Read file
$file = fopen("data.txt", "r");
$content = fread($file, filesize("data.txt"));
fclose($file);

echo $content;
?>
```

Output

Welcome to PHP File Handling

PHP File Inclusion

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- Include the content of a PHP file into another PHP file before the server executes it.
- There are two PHP functions which can be used to included one PHP file into another PHP file.
  - The include() Function
  - The require() Function

Include() example

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Want to create a common menu for your website. Then create a file menu.php with the following content.

```
<a href="http://www.PPSU.com/index.htm">Home</a>
<a href="http://www.PPSU.com/ebxml">ebXML</a>
<a href="http://www.PPSU.com/ajax">AJAX</a>
<a href="http://www.PPSU.com/perl">PERL</a>
```

```
<html>
   <body>

      <?php include("menu.php"); ?>
      <p>This is an example to show how to include PHP file!</p>

   </body>
</html>
```

It will produce the following result –

Home -
ebXML -
AJAX -
PERL

This is an example to show how to include PHP file!

EXAMPLE

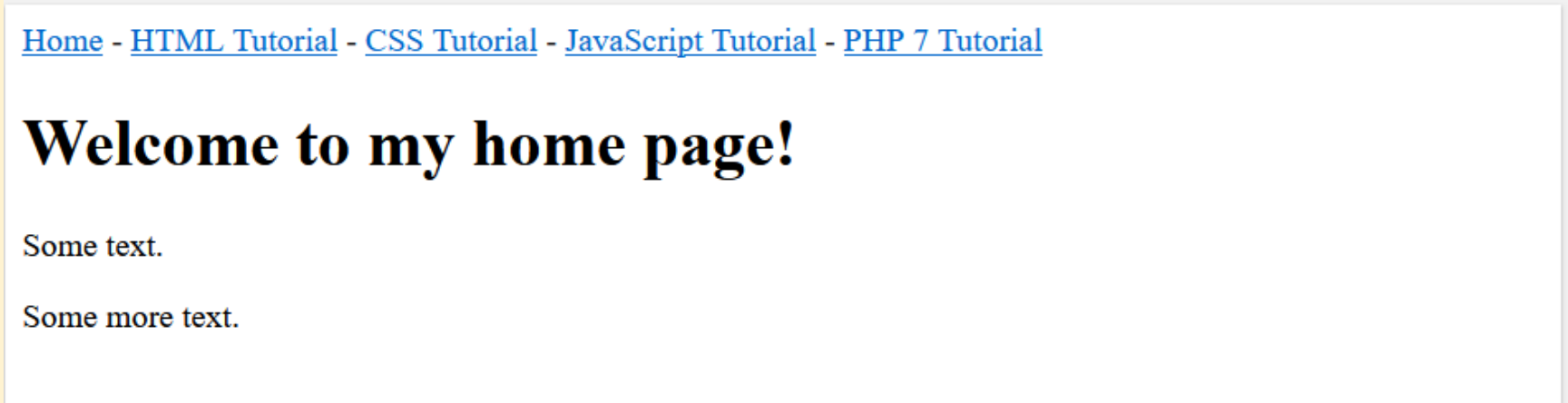
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```
<!DOCTYPE html>
<html>
<body>

<div class="menu">
<?php include 'menu.php';?>
</div>

<h1>Welcome to my home page!</h1>
<p>Some text.</p>
<p>Some more text.</p>

</body>
</html>
```



require() Function

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- It also includes the file in some another file.
- What's the difference with include()????
  - If there is any problem in loading a file then the **include()** function generates a warning but the script will continue execution.
  - If there is any problem in loading a file then the **require()** function generates a fatal error and halt the execution of the script.

EXAMPLE

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```
<!DOCTYPE html>
<html>
<body>

<h1>Welcome to my home page!</h1>
<?php include 'noFileExists.php';
echo "I have a $color $car.";
?>

</body>
</html>
```

Welcome to my home page!

I have a .

EXAMPLE

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- ```
  <!DOCTYPE html>
  <html>
  <body>

  <h1>Welcome to my home page!</h1>
  <?php require 'noFileExists.php';
  echo "I have a $color $car.";
  ?>

  </body>
  </html>
  ```

Welcome to my home page!
  ```

Include() vs. require()

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```
<html>
   <body>

      <?php include("xxmenu.php"); ?>
      <p>This is an example to show how to include wrong PHP file!</p>

   </body>
</html>
```

This will produce the following result –

```
This is an example to show how to include wrong PHP file!
```

Now lets try same example with require() function.

```
<html>
   <body>

      <?php require("xxmenu.php"); ?>
      <p>This is an example to show how to include wrong PHP file!</p>

   </body>
</html>
```

This time file execution halts and nothing is displayed.

What is a Cookie in PHP?

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- A **cookie** is a small piece of data stored on the **user's browser**.
- It is used to **remember user information** like username, theme, login status, etc.

- **Syntax to Set Cookie**
- setcookie(name, value, expiry_time);
- **name** → Cookie name
- **value** → Cookie value
- **expiry_time** → Time when cookie expires

Example: Set and Read Cookie

```
<?php
// Set cookie
setcookie("username", "Twisha", time() +
3600); // expires in 1 hour

// Read cookie
if(isset($_COOKIE["username"])) {
    echo "User Name: " .
$_COOKIE["username"];
} else {
    echo "Cookie is not set";
}
?>
```

Output

- **First time execution**

```
Cookie is not set
```

- **After refreshing the page**

```
User Name: Twisha
```

Reason: Cookies are available only after **page reload**.

Where Cookie is Stored?

- Stored in **client browser**
- Automatically sent to server with each request

Delete a Cookie

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```
<?php
setcookie("username", "", time() - 3600);
echo "Cookie deleted";
?>
```

Output

```
Cookie deleted
```

What is a Session in PHP?

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- A **session** is a way to store user data on the **server**.
- Sessions are used to keep data **available across multiple pages**, such as login information.

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- **Key Points about Sessions**
- Stored on **server side**
- More **secure** than cookies
- Session starts using session_start()
- Data stored using $_SESSION array

Example: Create and Read Session

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```
<?php
session_start();   // Start session

// Set session variable
$_SESSION["username"] = "Twisha
Patel";

// Read session variable
echo "Welcome " .
$_SESSION["username"];
?>
```

Output

```
Welcome Twisha Patel
```

Example: Access Session on Another Page



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Page 1: session_set.php

```
<?php
session_start();
$_SESSION["user"] = "Twisha";
echo "Session is set";
?>
```

Output

```
Session is set
```

Page 2: session_get.php

```
<?php
session_start();

if(isset($_SESSION["user"])) {
    echo "User Name: " .
$_SESSION["user"];
} else {
    echo "Session not found";
}
?>
```

Output

```
User Name: Twisha
```

Destroy Session

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```
<?php
session_start();
session_destroy();
echo "Session destroyed";
?>
```

Output

Session destroyed

Exercise

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1. Create a multidimensional array of 3 students with their marks.
2. Separate the marks for three subjects(Physics,Chemistry,Maths).
3. Display the marks of each student for each subject.
   - Using index
   - Using foreach loop
   - Using print_r method

Solution

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```php
<?php
$marks=array(
    "mani"=>array("physics"=>67,"maths"=>78),
    "deep"=>array("physics"=>87,"maths"=>79),
    "hari"=>array("physics"=>93,"maths"=>39)
);
//1st method
echo $marks["mani"]["physics"]."<br>";
echo $marks["mani"]["maths"]."<br>";
echo $marks["deep"]["physics"]."<br>";
echo $marks["deep"]["maths"]."<br>";
echo $marks["hari"]["physics"]."<br>";
echo $marks["hari"]["maths"]."<br>";
//2nd method
echo "<h4>printg by using foreach loop</h4>";
foreach($marks as $name=>$ary)
{
    foreach($ary as $subject=>$mark)
    {
        echo "name:$name, subject:$subject, marks:$mark<br>";
    }
}
```

Exercise

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1. Output "Hello World" if $a is greater than $b.
2. Output "Hello World" if $a is NOT equal to $b.
3. Output "Yes" if $a is equal to $b, otherwise output "No".
4. Print "1" if a is equal to b, print "2" if a is greater than b, otherwise print "3".

Exercise

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1. Create a switch statement that will output "Hello" if $color is "red", and "welcome" if $color is "green".
2. Add a section that will output "Neither" if $color is neither "red" nor "green".

- **Exercise:**

1. Write a function to calculate the factorial of a number (a non-negative integer).
2. Write a function to check whether a number is prime or not.
   Note: A prime number (or a prime) is a natural number greater than 1 that has no positive divisors other than 1 and itself.
3. Write a PHP function that checks whether a passed string is a palindrome or not?
   A palindrome is a word, phrase, or sequence that reads the same backward as forward, e.g., madam or nurses run.